

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Total Marks:50

QUESTIONS: 5

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

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I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.
5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 AT2 ANALYTICAL PROBLEM SOLVING.

Problem understanding:

We have a /16 block that must be divided using variable length subnet mask (VLSM) since each department needs a different prefix length. VLSM is the correct method here because fixed length subnetting would waste addresses - department have unequal size requirements.

Method:

Allocate subnets starting from the largest requirements down to the smallest assigning each block contiguously so that address ranges don't overlap. This standard VLSM best practice - allocate big to small to avoid fragmentation

HR Department - /25

Subnet : 172.16.0.0/25

Network Address : 172.16.0.0

Host Range : 172.16.0.1 - 172.16.0.126

Broadcast Address : 172.16.0.127

Usable hosts : $2^7 - 2 = 126$.

Sales Department - /26 :

Subnet : 172.16.0.128 /26

Network address : 172.16.0.128

Host range : 172.16.0.129

Broadcast address : 172.16.0.191

Usable host : $2^6 - 2 = 62$

IT department - /27 :

Subnet : 172.16.0.192 /27

Network address : 172.16.0.192

Host range : 172.16.0.193 - 172.16.0.222

Broadcast address : 172.16.0.223

Usable host : $2^5 - 2 = 30$

Admin department - /28 :

Subnet : 172.16.0.224 /28

Network address : 172.16.0.224

Host range : 172.16.0.225 - 172.16.0.238

Broadcast address : 172.16.0.239

Usable host : $2^4 - 2 = 14$

minimizing unused address range
total address consumed = $128 + 64 + 32 + 16 = 240$
(from 172.16.0.0 to 172.16.0.239)

unused range:

172.16.0.240 - 172.16.255.255

unused address count:

$65,536 - 240 = 65,296$ addresses available for future department or expansion.

Engineering relevance:

This VLSM allocation minimizes wastage compared to giving all departments equal size subnets, and leaves a large contiguous block for future growth.

4. Problem understanding:

A /24 network is being subdivided into fixed size /27 subnets. Since $/24 \rightarrow /27$ borrows 3 bits from the host portion this is a fixed length subnetting (FLSM) problem.

Method:

Number of subnets = $2^3 = 8$

Host bits remaining = $32 - 27 = 5 \rightarrow$ Addresses
per subnet = $2^5 = 32$.

Usable hosts per subnet = $2^5 - 2 = 30$.

#	Network address	usable host range	Broadcast address.
1	172.16.10.0/27	.1-30	172.16.10.31
2	172.16.10.32/27	.33-62	172.16.10.63
3	172.16.10.64/27	.65-94	172.16.10.95
4	172.16.10.96/27	.97-126	172.16.10.127
5	172.16.10.128/27	.129-158	172.16.10.159
6	172.16.10.160/27	.161-190	172.16.10.191
7	172.16.10.192/27	.193-222	172.16.10.223
8	172.16.10.224/27	.225-254	172.16.10.255

172.16.10.78

Converting the last octet (78) to binary: 01001110
With a /27 mask, the top 3 bits define the
subnet block (multiples of 32): 78 falls between
64 and 95.

Belongs to Subnet 3: 172.16.10.64/27

Network Address: 172.16.10.64

Broadcast address: 172.16.10.95

Checking 172.16.10.95:

This address is exactly the broadcast address
of Subnet 172.16.10.64/27

Yes 172.16.10.95 falls within the same subnet
block as 172.16.10.78 - but it is a broadcast address
not a usable host address. so both addresses
belong to same subnet 172.16.10.64/27.

Engineering relevance:

Assigning 172.16.10.95 to a host would
cause conflicts with the subnet's broadcast
function disrupting broadcast traffic delivery
in that subnet.